INCH-POUND

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30 JANUARY 1993
SUPERSEDING
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MILITARY SPECIFICATION

INDICATOR, POSITION, LANDING GEAR

This specification is approved for use by the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements for individual 28 volt direct current (dc) electrical landing gear position indicators.

1.2 Classification. The indicators shall be classified as follows:

MS27403-1	Lusterless White Dial Markings
MS27403-2	Fluorescent Luminescent Dial Markings

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

Federal

L-P-378	Plastic Sheet and Strip, Thin Gauge, Polyolefin
PPP-B-566	Box, Folding, Paperboard
PPP-B-636	Box, Shipping, Fiberboard

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/TICIA, Tinker AFB, OK 73145-5990 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6695

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MIL-P-116	Preservation, Method Of
MIL-C-5541	Chemical Films For Aluminum And Aluminum Alloys
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series; General Specification For
MIL-A-8625	Anodic Coatings, For Aluminum And Aluminum Alloys
MIL-T-31000	Technical Data Packages, General Specification For
MIL-C-83488	Coating, Aluminum, ION Vapor Deposited

STANDARDS

Federal

FED-STD-101	Preservation, Packaging And Packing Materials: Test Procedures
FED-STD-595	Colors

Military

MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking Of U.S. Military Property
MIL-STD-810C	Environmental Test Method And Engineering Guidelines
MIL-STD-889	Dissimilar Metals
MIL-STD-970	Specification And Standards, Order Of Precedence For The Selection Of
MIL-STD-2073/1B	Procedures For Development And Application Of Packaging Requirements
MS27403	Indicator, Position, Landing Gear, 28 Volt DC
MS33558	Numerals and Letters, Aircraft Instrument Dial, Standard Form
MS33678	Connectors, Receptacle, Electric, Integral Mounting

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Document Order Desk, Bldg 4D, 700 Robbins Ave, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

Air Force (AF) 8988994

(Copies of this AF drawing is available from OC-ALC/TILDOS, Tinker AFB OK 73145-5990.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Qualification. The indicator furnished under this specification shall be a product which has been tested, and passed the qualification tests specified herein and has been listed on, or approved for listing on the applicable qualified products list.

3.2 Selection of government specifications. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-970 except as provided in 3.2.1 and 3.2.2.

3.2.1 Commercial parts. Commercial parts having suitable properties may be used where, on the date of invitation for bids, there are no suitable standard parts. In any case, commercial utility parts, like screws, bolts, nuts, cotter pins, having suitable properties may be used provided:

a. They can be replaced by the standard parts (MS or AN) without alteration.

b. The corresponding standard part numbers are referenced in the parts list and if practical, on the contractors drawings.

2.2 Standard parts. With the exception of 3.2.1, MS and AN standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part numbers.

3.3 Materials. Materials shall conform to the applicable specifications and shall be specified herein. Materials for which there are no applicable specifications, or which are not specifically described herein, shall be of the best quality, of the lightest weight and suitable for the purpose intended.

3.3.1 Fungus proof materials. Materials which are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the contracting activity.

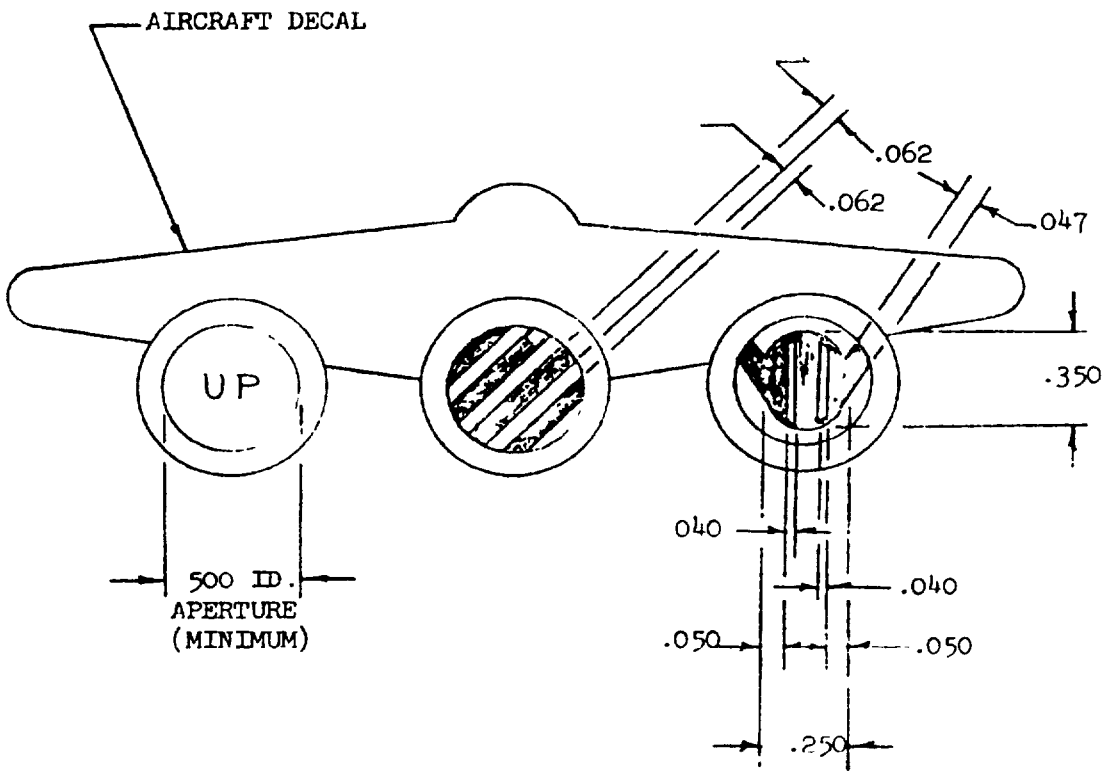
3.3.1.1 Reclaimed material. The use of reclaimed materials shall be encouraged to the maximum extent possible.

3.3.2 Metals. Metals shall be of a corrosion-resistant type or suitably treated to resist corrosion due to conditions likely to be encountered in storage or normal service life.

3.3.2.1 Dissimilar metals. Dissimilar metals as defined in MIL-STD-889 shall not be used in intimate contact with each other, unless suitable protected against electrolytic corrosion by means of protective coating.

3.3.3 Non-magnetic materials. Non-magnetic materials shall be used for all parts of the instrument except where magnetic materials are essential.

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Letters: 0.125 High; 0.0312 Line Width

Dimensions in inches: Tolerances ± 0.0156

FIGURE 1. Indications

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3.3.4 Corrosive fumes. The materials, as installed and under normal operating conditions, shall not liberate gases or fumes that will result in such corrosion as to cause malfunction of equipment or discoloration of dials.

3.4 Design and construction. The indicator shall be in accordance with MS27403, the case shall be of nonferrous, low density material and shall be finished in durable dull black, color No. 37038 of FED-STD-595. It shall be so constructed that no part will work loose in service and shall be built to withstand the strains, jars, vibration and other conditions incident to shipping, storage, installation and service without failure.

3.4.1 Dial indication. The indicator shall indicate the landing gear position by means of moveable tabs. The dimensions and appearance of the tabs shall conform to the three position indications shown in Figure 1. The wheel image shall be positioned at the top of the dial drum and the letters "UP" shall be positioned at the bottom of the dial drum.

3.4.2 Power. The indicator shall be designed for operation on 28 volts dc. The current required for each operating circuit shall not exceed 0.03 amp. Resistors shall not be used to control power consumption.

3.4.3 Wiring. The indicator shall be so wired that the receptacle B terminal is common and positive. The circuit through terminal A shall activate the "UP" tab and terminal C the wheel tab.

3.4.4 Electrical receptacle. The receptacle shall be suitable protected from corrosion. The receptacle shall be located as shown on MS27403, except that it may be moved not more than 0.1875 inch off the centerline of the indicator.

3.4.5 Reinforcement. The inside of the case around each connection shall be of sufficient strength to prevent damage to the case when the connections are tightened during installatio of the indicator.

3.4.6 Hermetically sealed case. The case shall provide a hermetically sealed enclosure for the internal mechanism. The hermetic sealing shall be so accomplished that the seal will not be dependent upon materials that will be affected by the action of any atmosphere to which the indicator may be subjected.

3.4.6.1 Helium pressure. The case shall be filled with helium of at least 98 percent purity, free of dust particles, and containing no more than 0.006 milligrams (mg) of water vapor per liter (dew point minus 65o Centigrade (C) at the filling pressure. The absolute pressure of the helium within the case shall be approximately 1 atmosphere.

3.4.7 Marking. The top side of the case shall be marked "THIS END UP" to facilitate proper mounting of the indicator. The marking "THIS END UP", with an arrow pointing toward the top of the case, may be placed on each side of the case in liue of the top.

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3.4.8 Cover glass. The cover glass shall be clear and free from flaws which interfere with normal reading of the indicator. The thickness of the glass shall be entirely suitable for the purpose and shall not exceed 0.047 inch. The distance between the inside surface of the cover glass and the arc of the dial shall not exceed 0.0312 inch.

3.5 Performance. The indicator shall be capable of satisfactory operation when subjected to the following conditions:

a. Temperature. Temperatures ranging from minus 54° C to plus 71° C.

b. Salt fog. Exposure to salt sea atmosphere.

c. Vibration. Vibration incident to service use.

d. Shock. Shock forces of 100g.

e. Thermal shock. Thermal shock as specified in 4.

f. Sealing. Immersion in a deaerated liquid without leaking.

3.6 Interchangeability. All parts having the same manufacturers part number shall be functionally and dimensionally interchangeable. The drawing number requirements of MIL-T-31000 shall govern changes in the manufacturers part numbers.

3.7 Markings.

3.7.1 Dial markings. All markings shall be durable to withstand the usage encountered in service. The style and proportions of numerals and letters placed on the dial shall conform to MS33558.

3.7.2 Dial marking material. As specified in the contract or purchase order, the following markings shall be fininshed in fluorescent-luminescent material in accordance with AF8988994, Type I or III as applicable, or in lusterless white material, color number 37875 of FED-STD-595.

3.7.3 Lusterless black. The background of the dial and all other markings not otherwise specified shall be finished in lusterless black, color number 37038 of FED-STD-595.

3.8 Weight. The weight of the completely assembled indicator, less mounting screws, shall not exceed 2.5 ounces.

3.9 Screw threads. Unless otherwise specified, the threads of all machine screws 0.060 inch or larger in diameter, shall conform to MIL-S-7742.

3.10 Finishes and protective coatings.

3.10.1 Aluminum alloy patrs. Alumimum alloy parts shall be covered with an anodic film conforming to MIL-A-8625, except as follows:

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a. Dials, small holes and case inserts need not be anodized.

b. Aluminum alloys which do not anodize satisfactorily shall be coated with a chemical film IAW MIL-C-5541.

c. Where a primary purpose of the treatment is to provide a suitable paint base, chemical treatments IAW MIL-C-5541 may be used in lieu of anodizing.

d. Castings containing non-aluminum alloy integral inserts may be treated with a chemical film in accordance with MIL-C-5541 in lieu of anodizing.

e. When abrasion resistance is a factor, chemical films in accordance with MIL-C-5541 shall not be used in lieu of anodizing.

f. Parts contained within hermetically sealed enclosures are considered to be suitably protected without anodizing or plating.

3.10.2 Steel parts. Steel parts shall be coated with ion vapor deposited aluminum, where practicable, in accordance with MIL-C-83488, type I or II as applicable and of a class that is adequate to achieve the degree of protection required. Other protective coating, in lieu of MIL-C-83488, may be used if demonstrated to be satisfactory and approved by the preparing activity. Cadmium plating must be avoided when satisfactory alternative processes can be used.

3.11 Identification of product. A nameplate shall be securely attached to the outside of the case and shall be legibly and duably marked in accordance with MIL-STD-130.

3.12 Workmanship. The indicator, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, marking of parts and assemblies, and freedom of parts from burrs and sharp edges.

3.13 Installation.

3.13.1 Mounting screws. The contractor shall furnish the required mounting screws and lockwashers to install the indicator. Screws shall be number 6-32 round head, brass machine screws, NC-2, with a durable dull black finish and of proper length to permit mounting on a panel 0.125 inch thick.

3.13.2 Envelope. An envelope containing the required screws and lockwashers shall be furnished with each indicator. The face of the envelope shall be marked to indicate the contents.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance

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of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.4).

b. Quality conformance inspection (see 4.7).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in 4.3.1.

4.3.1 Standard atmospheric conditions. Whenever the pressure and temperature are not specified, it is understood that the test will be made at atmospheric pressure (approximately 29.92 inchge Hg) and at room temperature (approximately plus 25° C). When tests are made with atmoshperic pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified condition.

4.3.2 Attitude. Unless otherwise specified, the indicator shall be tested in the normal operating position.

4.3.3 Vibration. A vibration stand which will vibrate at any desired frequency between 300 and 3000 cycles per minute shall be used to subject the indicator to vibration such that a point on the case will describe in a plane inclined 45° to the horizontal plane, a circle, the diameter of which is equal to the amplitude specified herein.

4.3.4 Voltage input. Unless otherwise specified, the indicator shall be tested with 28 volts dc.

4.4 Qualification inspection.

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4.4.1 Test samples. The test samples shall consist of two indicators. Samples shall be identified with the manufacturers own part number and any additional information required by the letter of authorization.

4.4.2 Qualification required. Prior to actual procurement, the product which this specification covers shall pass the qualification tests specified herein.

4.4.3 Qualification inspection. The qualification inspection shall consist of all the tests of this specification.

4.5 Acceptance tests. Acceptance tests shall consist of:

a. Individual tests.

b. Sampling plans and tests.

4.5.1 Individual tests. Each indicator shall be subjected to the following tests as described under 4.6:

a. Examination of product.

b. Operation.

c. Friction.

d. Indicator position.

e. Dielectric strength.

f. Sealing.

4.5.2 Sampling plans and tests.

4.5.2.1 Sampling plan A. Five indicators selected at random from every 500 or less on the contract or order shall be subjected to the following tests as described under 4.6:

a. Low temperature operation.

b. High temperature operation.

c. Magnetic effect.

d. Vibration error.

e. Tab variation.

f. Shock tests.

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4.5.2.2 Sampling plan B. Unless otherwise specified, 3 indicators selected at random from the first 15 items of the contract or order shall be subjected to the following tests as described under 4:

a. Low temperature exposure.

b. High temperature exposure.

c. Voltage variation.

d. Indicator life.

e. Vibration failure.

f. Salt fog.

g. Thermal shock.

4.5.2.2.1 Rejection and retest. When 1 item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure are determined.

4.5.2.2.1.1 Individual tests may continue. For operational reasons, individual tests may be continued pending the investigation of a sampling test failure. Final acceptance of items on hand or later produced shall not be made until it is determined that items meet all the requirements of the specification.

4.5.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.6 Test methods.

4.6.1 Examination of product. Each indicator shall be inspected to determine conformance with this specification with respect to material, workmanship, and envelope.

4.6.2 Operation. The indicator shall be connected to a 28 volt dc source and a single-pole, double-throw switch or an equivalent circuit. The B terminal shall be positive and common. The circuit shall be completed alternately through terminals A and C, and th "UP" tab and wheel tab shall appear respectively. Whenever current is not applied to either terminal, the unlocked tab shall appear as shown in Figure 1.

4.6.3 Friction. The indicator shall be operated as above, except that the indication before and after tapping shall be noted. The indicator shall be in the normal upright mounting position and shall be tapped only after the tab has come to rest. The difference in indication before and after tapping shall not exceed 1/32 inch of tab movement.

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4.6.4 Indicator position. The indication shall be noted with the indicator energized in the "UP" position, unlocked, and wheel indications for the following positions:

a. Normal upright dial vertical position.

b. Tipped 90° rearward, the dial horizontal and face up.

c. Tipped 90° to the right, the dial in a vertical plane.

d. Tipped 90° to the left, the dial in a vertical plane.

The maximum deviation of reading (b), (c), and (d) shall not exceed 1/32 inch of tab movement.

4.6.5 Dielectric strength. The application of 200 volts, 60 cycles per second (cps) for a 5 second interval between the individual pins and the indicator case shall cause no evidence of insulation breakdown.

4.6.6 Low temperature operation. The indicator shall be subjected to a temperature of 540 + 2° C for a period of 4 hours. At the end of this period and while still minus 54° C, the indicator shall be subjected to, and shall meet the requirements of, the tests specified in 4.6.2 and 4.6.3. After the indicator has returned to room temperature, it shall be subjected to and shall meet the requirements of the tests specified in 4.6.2 and 4.6.3.

4.6.7 High temperature operation. The indicator shall be subjected to a temperature of 71° + 2° C for a period of 4 hours. At the end of this period and while still at minus 71° C, the indicator shall be subjected to and shall meet the requirements of, the tests specified in 4.6.2 and 4.6.3. After the indicator has returned to room temperature, it shall be subjected to and shall meet the requirements of the tests specified in 4.6.2 and 4.6.3.

4.6.8 Magnetic effect. The indicator, first not operating, and second, operating on rated voltage, shall be rotated in a vertical plane about a short bar magnet compass with the nearest part of the indicator 5 1/2 inches from the magnetically East or West of the center of the compass. Starting directly under the compass, the indicator shall be held in position s 0°, 45°, 90°, 135°, 180°, 225°, 270°, and 315° from the initial position. At each of these positions, the indicator shall be rotated on its own horizontal axis until it is in its normal position. The horizontal magnetic field intensity shall be 0.17 to 0.19 oersted. The deflection of the compass at any of the specified positions shall not exceed 3°.

4.6.9 Vibration error. The indicator shall be mounted on the vibration stand in its normal operating position. The indicator shall be operated on 29 volts dc and shall be subjected to vibration from 5 to 50 cycles per second with an amplitude of 0.009 to 0.011 inch. The maximum amplitude of the moveable tab vibration and the tab variation shall not exceed 1/32 inch when the indication is in either the "UP" or "DOWN" position.

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4.6.10 Shock test. Method 516.2, Procedure I of MIL-STD-810C. The test specimen shall be subjected to 18 impact shocks of 100g, each shock impulse having a time duration of 7.5 + 1 milliseconds. The intensity shall be within + 10 percent when measured with a filter having a bandwidth of 5 to 100 cps, the maximum g shall be reached in approximately 3.75 milliseconds. The shock shall be applied in the following directions:

a. Vertical axis, 3 shocks in each direction.

b. Horizontal axis (major and minor) 3 shocks in each direction.

This test shall be accomplished on 3 indicators with the power off and 1 indicator with pins A and B energized and on 1 indicator with pins C and B energized. Upon completion of this test, the indicators shall be tested and must comply with all of the tests specified in 4.5.1.

4.6.11 Low temperature exposure. The indicator shall be subjected to a temperature of minus $64^{\circ} + 2^{\circ}$ C for a period of 48 hours. While still at this temperaturem the indicator shall be operated throughout its range without exhibiting sticking or erratic characteristics. The temperature shall then be raised to minus $54^{\circ} + 2^{\circ}$ C for a period of 24 hours. At the end of this period and while still at minus 54° C, the indicator shall be subjected to and shall meet the requirements of the tests specified in 4.6.2 and 4.6.3. Upon return to room temperature, the indicator shall be subjected to and shall meet the requirements of the tests specified in 4.6.2 and 4.6.3.

4.6.12 High temperature exposure. The indicator shall be subjected to a temperature of $71^{\circ} + 2^{\circ}$ C for a period of 24 hours. At the end of this period and while still at 71° C, the indicator shall be subjected to and shall meet the requirements of the tests specified in 4.6.2 and 4.6.3. Upon return to room temperature, the indicator shall be subjected to and shall meet the requirements of the tests specified in 4.6.2 and 4.6.3.

4.6.13 Voltage variation. The indicator shall be operated at a reduced voltage of 22 1/2 volts and an increased voltage of 30 volts. There shall not be more than 1/32 inch movement of the tab between operation at the reduced voltage and operation at the increased voltage. The indicator shall be tapped lightly after each change in voltage.

4.6.14 Indicator life. The indicator shall be operated for 20,000 cycles at a rate of 10 to 12 cycles per minute. A cycle shall consist of movement from the unlocked position to the "UP" locked position, to the unlocked position, to the wheel "DOWN" position and bact to the unlocked position. Upon completion of the test, the indicator shall be subjected to and shall meet the requirements of the tests specified in 4.6.2 and 4.6.3.

4.6.15 Vibration failure. The indicator shall be subjected to a vibration test in accordance with Method 514.2, Procedure I, Part I, utilizing Table 514.2-II and Figure 514/2-2, Curve C, Category b.1 of MIL-STD-810C. The test shall last for a period of three hours. After vibration, the indicator shall be subjected to and shall meet the requirements of the tests specified in 4.6.2 and 4.6.3.

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4.6.16 Salt fog. The indicator shall be subjected to salt fog in accordance with MIL-STD-810C, Method 509.1, for 50 hours. At the end of this period, the indicator shall be examined. There shall be no corrosion or resultant damage from this test. The indicator shall then be subjected to and shall meet the requirements of the tests specified in 4.6.2, 4.6.3 and 4.6.5.

4.6.17 Sealing. The indicator shall be subjected to Immersion tests, Method 512.1, Procedures III of MIL-STD-810C. The absolute pressure of the air above liquid shall be reduced one inch of mercury for one minute, then increased 2.5 inches for one hour.

4.6.18 Thermal shock. The indicator shall be subjected to 8 cycles of exposure to water at $85^{o} + 20$ C and $5^{o} + 2^{o}$ C without evidence of moisture penetration or damge to coating or enclosure. Each cycle of the test shall consist of immersing the indicator shall be immersed in the 5° C water for a period of 25 minutes. This cycle shall be repeated continuously until 8 cycles have been completed. Following this test, the sealing test shall be repeated. No indicator leakage shall occur as a result of thermal shock.

4.7 Packaging inspection.

4.7.1 Quality conformance. The inspection of the preservation-packaging and interior package marking shall be in accordance with group A and B quality conformance inspection requirements, section 4 of MIL-STD-116. The sampling and inspection of the quality assurance provisions of applicable container specification and the marking requirements of MIL-STD-129.

4.7.2 First article. When the unit container is capable of serving as the shipping container, First Article inspection and rough handling tests, as outlined in section 4 of MIL-P-116, shalll be accomplished followed by a functional test of the unit to ensure freedom from operational malfunction.

5. PACKAGING.

5.1 Preservation. Preservation shall be level "A" or "C", as specified. (See 6.2)

5.1.1 Preservation-packaging. Unless otherwise specified by the contracting activity, item shall be packaged in quantity unit pack (QUP) of one each and intermediate container quantity (ICQ) of fifty (50) each. Each item will be provided a preservation method IC-1 IAW MIL-P-116.

5.1.2 Cleaning. Item shall be cleaned IAW MIL-P-116 C-1 process.

5.1.3 Drying. Immediately after cleaning, the item shall be dried following any one or combination of the drying procedures listed in MIL-P-116. The drying procedures employed shall not be injurious to the item.

5.1.4 Level A. Item shall be preserved IAW MIL-P-116 and MIL-STD-2073-1B to provide a method IC-1.

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5.1.5 Level C. Item shall be individually preserved in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from the supply source to the first receiving activity as specified in MIL-P-116 and MIL-STD-2073-1B.

5.2 Packing. Packing shall be level "A", "B", or "C". as specified. (See 6.2)

5.2.1 Container-cushioning. Unit container requirements shall conform to PPP-B-556, weather-resistant, unless otherwise specified by the contracting activity. Unit containers shall be large enough to allow for application of sufficient cushioning material, between the item and unit container. Cushioning material shall be of sufficient density and thickness to protect the item from impact shock damage. Intermediate container requirements shall conform to PPP-B-636, weather-resistance unless otherwise specified by the contracting activity. Intermediate container shall be large enough to accommodate fifty (50) unit containers.

5.2.2 Level A. Item will be preserved as specified in 5.1 and shall be packed in exterior containers conforming to PPP-B-636, weather-resistant, unless otherwise specified by the contracting activity. Exterior container shall be uniform shape, size and minimum tare and cube, consistent with the protection required. Closure shall be IAW appropriate PPP-B-636 procedure, as specified by contractor, special packaging instructions (SPI), and contracting activity.

5.2.3 Level B. Same as Level "A" unless otherwise specified by SPI or the contracting activity.

5.2.4 Level C. Item will be packed in such a manner that will afford adequate protection against physical damage during direct domestic shipment from the supply source to the first receiving activity. These packs shall conform to MIL-STD-2073-1B.

5.3 Marking. Unit, intermediate, and exterior containers shall be marked IAW special markings required by the contractor, SPI and MIL-STD-129.

5.4 Inspection and test. Test of methods of preservation shall be accomplished IAW section 4 of MIL-P-116 to ensure compliance with section 5 of this specification. Packaging tests shall be conducted IAW rough handling as specified in FED-STD-101.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but not mandatory.)

6.1 Intended use. The individual landing gear position indicator covered by this specification is intended for use on all aircraft with retractable landing gear where indication of the position of the landing gear is desired.

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6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitaion, and if required, the specific issue of individual documents referenced. (See 2.1)

c. The quantity and MS part number of the indicator desired.

d. Selection of applicable levels of packaging and packing.

e. The type of material to be used in marking the dial.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Oklahoma City Air Logistics Center/TICLA, Tinker AFB, OK 73145-5990 and information pertaining to qualification of products may be obtained from that activity.

6.4 Definitions.

6.4.1 Amplitude. Whenever the word "amplitude" is specified, it shall mean the extent of motion as measured from one extreme to the opposite extreme.

6.4.2 Hermetic seal. A hermetic seal is defined as a perfectly closed and airtight seal made between vitric, metallic, elastomeric or resinous materials.

6.5 Subject term (key word) listing.

Electrical landing gear
Position indicator
28 volt

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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Preparing Activity:
AIR FORCE -71

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I RECOMMEND A CHANGE:	1 DOCUMENT NUMBER MIL-I6839C	2 DOCUMENT DATE (YYMMDD) 93/01/30

3 DOCUMENT TITLE
INDICATOR, POSITION, LANDING GEAR

4 NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible Attach extra sheets as needed)

5 REASON FOR RECOMMENDATION

6. SUBMITTER

a NAME (Last, First, Middle Initial)	b ORGANIZATION	
c. ADDRESS (Include Zip Code)	d TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

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